

	1	2	3	4	5	6
A	Power input, USB-C and 3.3 V File: 01_Power_Input_USB.kicad_sch		ESP32-S3, native USB, reset and boot File: 02_ESP32_USB_Reset.kicad_sch			A
B	E-paper power and interface File: 03_Display_EPaper.kicad_sch		Touch always-on rail and RTC File: 04_Touch_RTC.kicad_sch			B
C	AUTOTERM UART and 1-Wire File: 05_AUTOTERM_1Wire.kicad_sch		External controls, lighting and diagnostics File: 06_Controls_Diagnostics.kicad_sch			C
D					ESP32-S3-WROOM-1U-N16R8 / 10-14V vehicle auxiliary battery / USB-C Review schematic - NOT RELEASED FOR PRODUCTION LandyHeater Sheet: / File: LandyHeater-Board.kicad_sch Title: Landy Heater Controller - Hierarchy Size: A4 Date: 2026-09-06 Rev: A / Phase 2 KiCad E.D.A. 10.0.6 Id: 1/7	D
	1	2	3	4	5	6

12-V INPUT / REVERSE BATTERY / OV CUTOFF

FLOW: J1 -> TVS + LM74720/Q1 -> FILTER -> VIN_SYS

TP1 TP_BATT_12V

TP3 TP_VIN_SYS

TP4 TP_3V3

TP5 TP_GND

USB-C UFP / CURRENT DETECTION / TRUE REVERSE BLOCKING

FLOW: J5 -> TUSB321 + TPS259470 -> POWER OR -> LMR43620/L2 -> 3V3_CORE

12 V protected input | USB-C sink and reverse blocking | 3.3 V automotive buck
Review schematic – NOT RELEASED FOR PRODUCTION
LandyHeater
Sheet: /Power input, USB-C and 3.3 V/
File: 01_Power_Input_USB.kicad_sch

Title: Power input, USB-C and 3.3 V

Size: A3	Date: 2026-09-06	Rev: A / Phase 2
KiCad E.D.A. 10.0.6		Id: 2/7

12-V INPUT / REVERSE BATTERY / OV CUTOFF

FLOW: J1 -> TVS + LM74720/Q1 -> FILTER -> VIN_SYS

TP1 TP_BATT_12V

TP3 TP_VIN_SYS

TP4 TP_3V3

TP5 TP_GND

USB-C UFP / CURRENT DETECTION / TRUE REVERSE BLOCKING

FLOW: J5 -> TUSB321 + TPS259470 -> POWER OR -> LMR43620/L2 -> 3V3_CORE

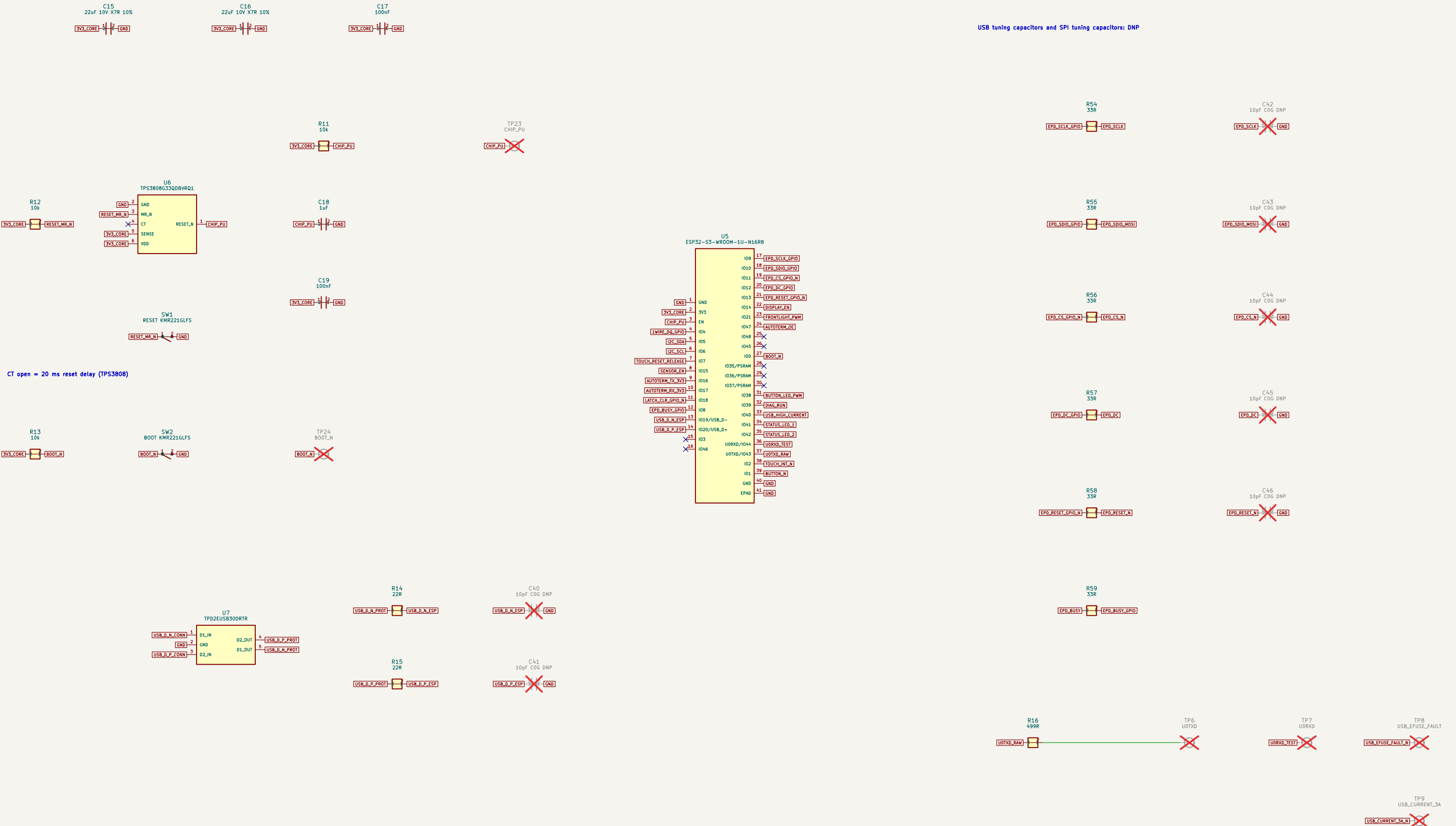
12 V protected input | USB-C sink and reverse blocking | 3.3 V automotive buck
Review schematic – NOT RELEASED FOR PRODUCTION
LandyHeater
Sheet: /Power input, USB-C and 3.3 V/
File: 01_Power_Input_USB.kicad_sch

Title: Power input, USB-C and 3.3 V

Size: A3	Date: 2026-09-06	Rev: A / Phase 2
KiCad E.D.A. 10.0.6		Id: 2/7

ESP32-S3-WROOM-1U-N16R8 / SAFE RESET AND BOOT

FLOW: 3V3 -> TPS3808/RESET -> CHIP_PU | USB J5 -> ESD -> R14/R15 -> GPIO19/20 | GPIO -> SPI SERIES PARTS



CT open = 20 ms reset delay (TPS3808)

N16R8 octal-PSRAM pins reserved | Native USB GPIO19/20 | Manual BOOT and RESET
Review schematic – NOT RELEASED FOR PRODUCTION
LandyHeater

Sheet: /ESP32-S3, native USB, reset and boot/
File: 02_ESP32_USB_Reset.kicad_sch

Title: *ESP32-S3, native USB, reset and boot*

Size: A3	Date: 2026-09-06
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Rev: A / Phase 2

KiCad E.D.A. 10.0.6

Id: 3/7

DISPLAY POWER-OFF ISOLATION

GOOD DISPLAY CHAPTER 12 BOOSTER – DO NOT SUBSTITUTE TOPOLOGY

FLOW: GPIO SPI -> AXC LEVEL/POWER-OFF ISOLATION -> J6 | 3V3_DISPLAY_SW -> BOOSTER -> PANEL RAILS

A

B

C

D

E

F

A

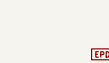
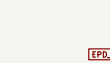
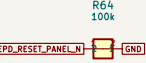
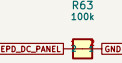
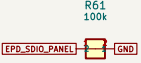
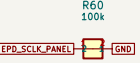
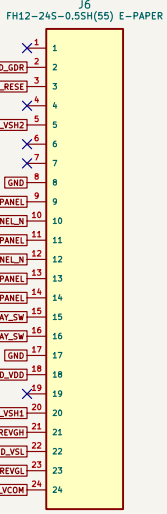
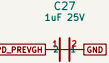
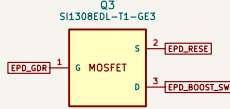
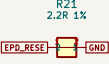
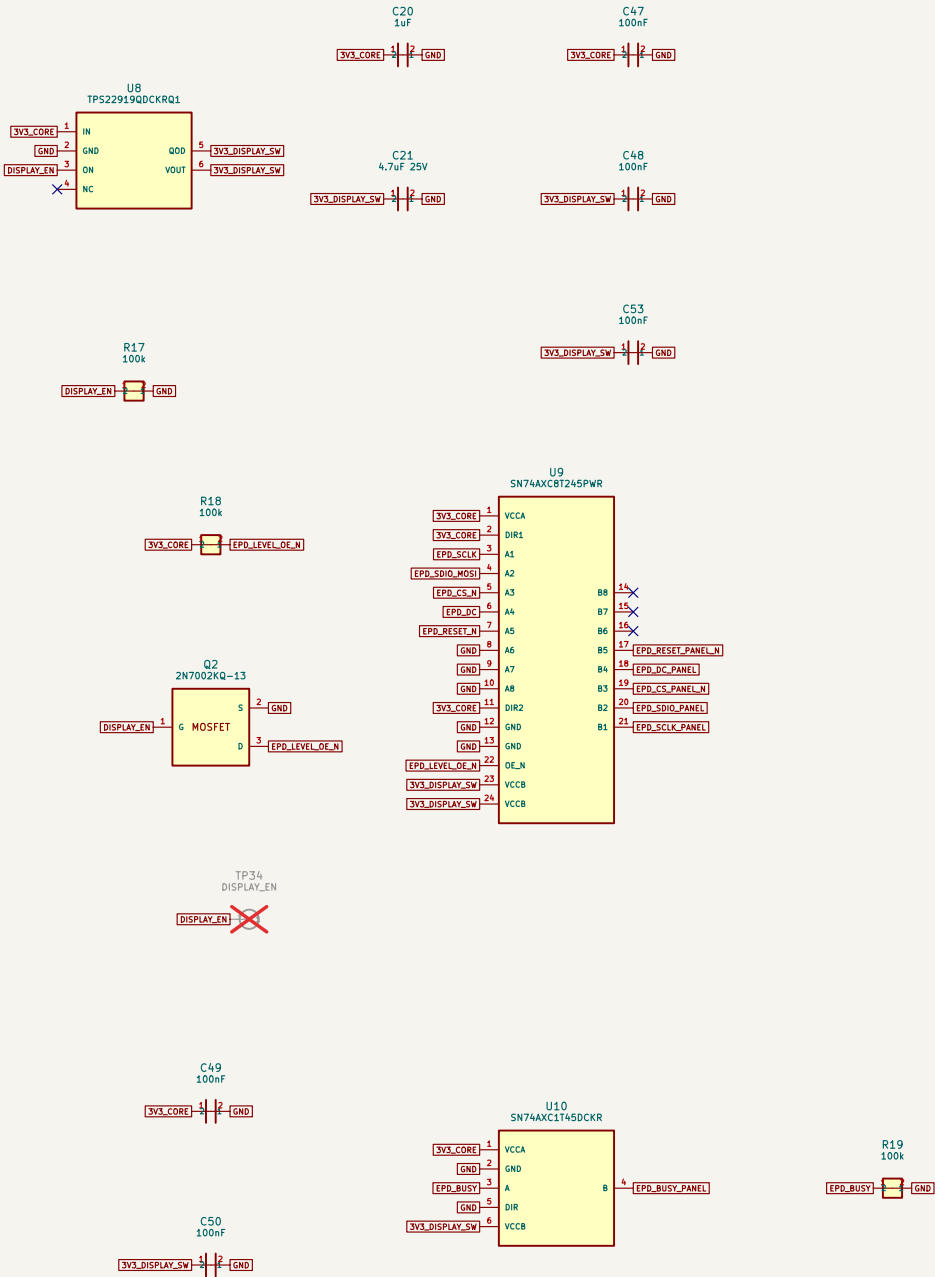
B

C

D

E

F



FPC contact side / pin-1 orientation: NOT VERIFIED until original sample inspection

Write-only SPI | Power-off isolation | FPC orientation remains a physical sample gate
Review schematic – NOT RELEASED FOR PRODUCTION

LandyHeater

Sheet: /E-paper power and interface/
File: 03_Display_EPaper.kicad_sch

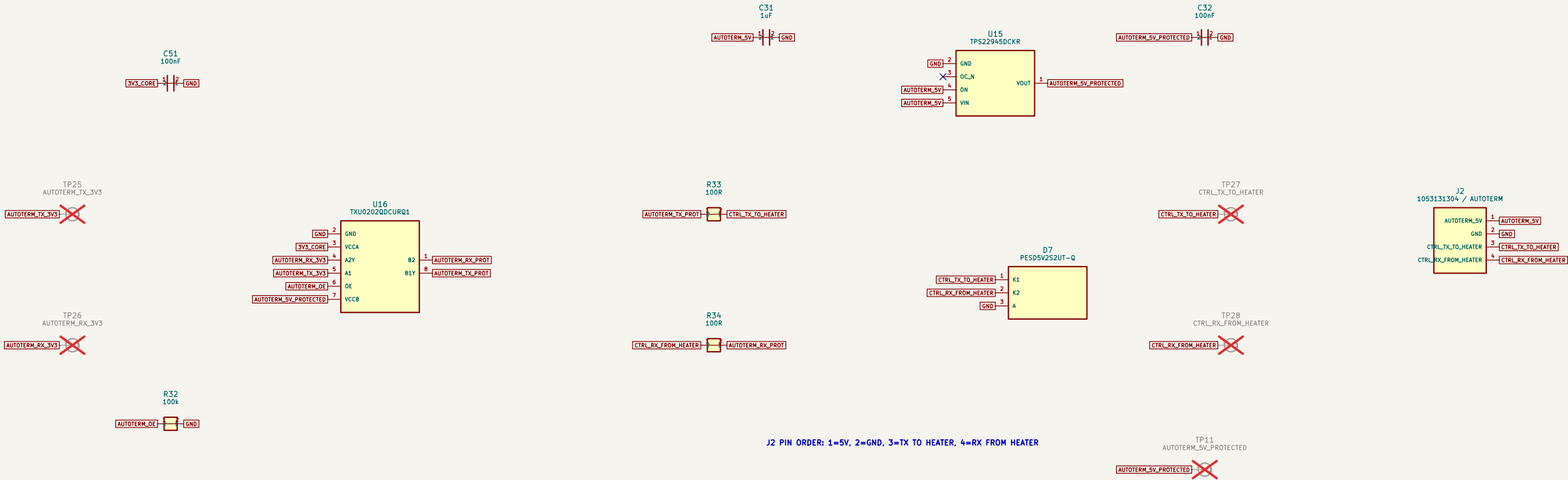
Title: E-paper power and interface

Size: A3 Date: 2026-09-06
KiCad E.D.A. 10.0.6

Rev: A / Phase 2
Id: 4/7

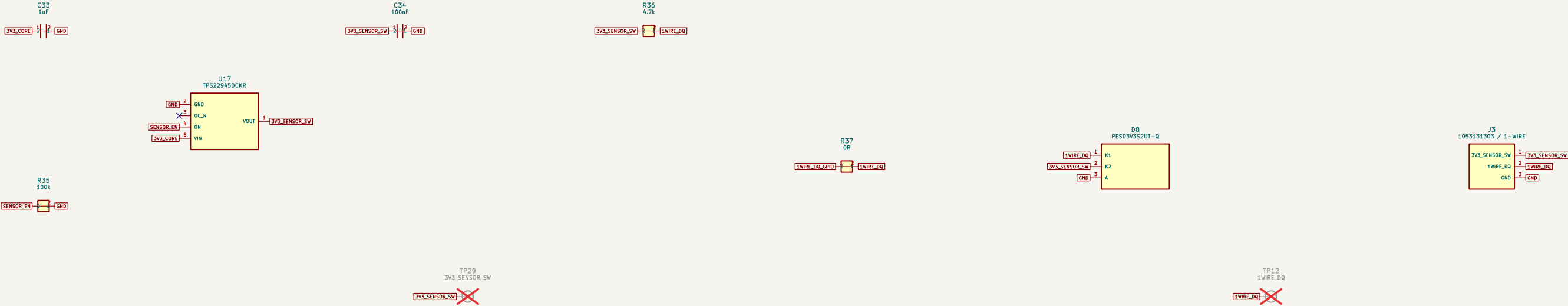
AUTOTERM UART – 5–V side powered only by heater

FLOW: ESP32 UART -> TXU0202 -> SERIES + ESD -> J2



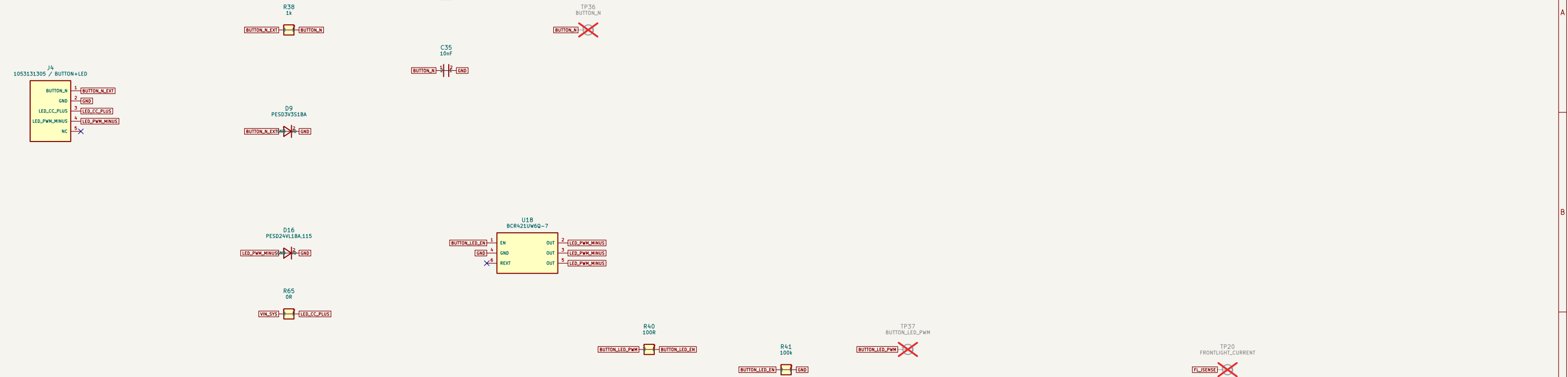
1–WIRE – THREE DS18B20 / 2-5 m combined harness

FLOW: 3V3 -> TPS22945 -> SENSOR SUPPLY | GPIO4 -> OR + ESD -> J3



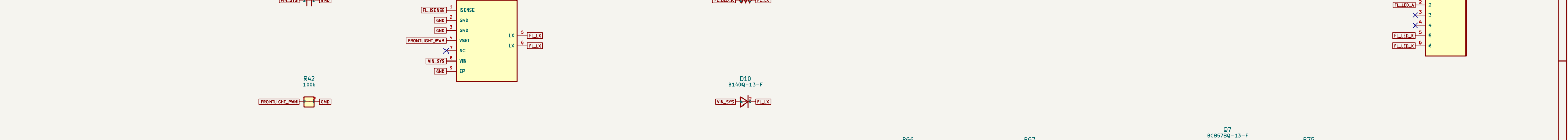
EXTERNAL BUTTON + WHITE RING LED / 3 m

FLOW: J4 BUTTON -> FILTER/ESD -> GPIO1 | GPIO38 -> BCR421 PWM -> J4 RING LED

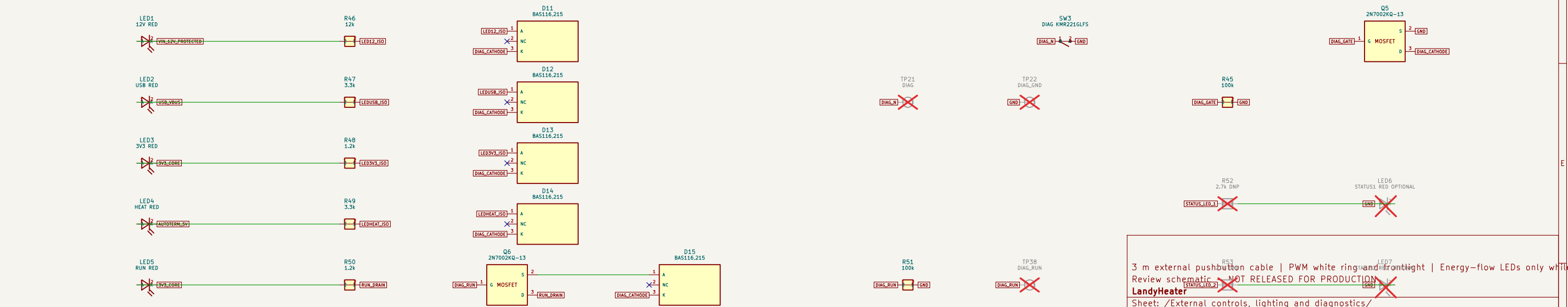


FRONTLIGHT 50 mA NOMINAL / <60 mA WORST CASE

FLOW: VIN_SYS -> AL8861/L4/D10 -> J8 FRONTLIGHT



ENERGY-FLOW DIAGNOSTICS – LEDs only while DIAG is pressed



3 m external pushbutton cable | PWM white ring and frontlight | Energy-flow LEDs only while DIAG is pressed

Review schematic - NOT RELEASED FOR PRODUCTION

LandyHeater

Sheet: /External controls, lighting and diagnostics/
File: 06_Controls_Diagnostics.kicad_sch

Title: External controls, lighting and diagnostics

Size: A3

Date: 2026-09-06

Rev: A / Phase 2

KiCad E.D.A. 10.0.6

Id: 7/7